F U N C T I O N S

Quick Revision Formula Sheet

Function

A relation:

$$f : A \to B$$

where every element of A has exactly one image in B.

$$y = f(x)$$

Domain

Set of all possible values of x.

Example:

$$f(x) = \frac{1}{x-2}$$

Domain:

$$x \neq 2$$

Range

Set of all possible values of y.

Example:

$$f(x) = x^2$$

Range:

$$y \geq 0$$

Codomain

Target set in mapping:

$$f : A \to B$$

Here B is codomain.

One-One Function

$$f(a) = f(b) \Rightarrow a = b$$

Distinct inputs give distinct outputs.

Many-One Function

Different inputs may give same output.

Example:

$$f(x) = x^2$$

$$f(2) = f(-2) = 4$$

Onto Function

Every element of codomain has pre-image.

$$\text{Range} = \text{Codomain}$$

Into Function

Some elements of codomain have no pre-image.

$$\text{Range} \subset \text{Codomain}$$

Bijective Function

Both:

- One-One
- Onto

Identity Function

$$f(x) = x$$

Every element maps to itself.

Constant Function

$$f(x) = c$$

Output remains constant.

Polynomial Function

$$f(x) = a_n x^n + \cdots + a_1 x + a_0$$

Linear Function

$$f(x) = ax + b$$

Quadratic Function

$$f(x) = ax^2 + bx + c$$

Modulus Function

$$f(x) = |x|$$

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Greatest Integer Function

$$f(x) = [x]$$

Greatest integer less than or equal to x.

Signum Function

$$\operatorname{sgn}(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$

Even Function

$$f(-x) = f(x)$$

Symmetric about y-axis.
Example:

$$x^2,\ \cos x$$

Odd Function

$$f(-x) = -f(x)$$

Symmetric about origin.
Example:

$$x^3,\ \sin x$$

Composite Function

$$(f \circ g)(x) = f(g(x))$$

Inverse Function

$$f^{-1}(f(x)) = x$$

Exists only for bijective functions.

Exponential Function

$$f(x) = a^x$$

where:

$$a > 0, \quad a \neq 1$$

Logarithmic Function

$$f(x) = \log_a x$$

$$a > 0, \quad a \neq 1$$

Trigonometric Functions

$$\sin x, \quad \cos x, \quad \tan x$$

Important Domains

$$\sqrt{x} \Rightarrow x \geq 0$$

$$\frac{1}{x} \Rightarrow x \neq 0$$

$$\log x \Rightarrow x > 0$$

FORMULA SHEET

Composition Property

If:

$$f : A \to B$$

$$g : B \to C$$

then:

$$g \circ f : A \to C$$

Graph Symmetry

Even Function:

Y-axis symmetry

Odd Function:

Origin symmetry

Quick Identities

$$f^{-1}(x) = \frac{1}{x}$$

for:

$$f(x) = x$$

$$(a^x)' = a^x \ln a$$

$$(\log_a x)' = \frac{1}{x \ln a}$$

Function Operations

Addition:

$$(f + g)(x) = f(x) + g(x)$$

Subtraction:

$$(f - g)(x) = f(x) - g(x)$$

Multiplication:

$$(fg)(x) = f(x)g(x)$$

Division:

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

Important Graphs

$$y = x$$

Straight line

$$y = x^2$$

Parabola

$$y = |x|$$

V-shape graph

$$y = \sin x$$

Wave graph

Limit

$$\boxed{\lim_{x \to a} f(x) = L}$$

means:

$$f(x) \to L \quad \text{as} \quad x \to a$$

Left Hand Limit

$$\boxed{\lim_{x \to a^-} f(x)}$$

Value approached from left side.

Right Hand Limit

$$\boxed{\lim_{x \to a^+} f(x)}$$

Value approached from right side.

Condition for Existence of Limit

Limit exists if:

$$\boxed{\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x)}$$

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Basic Laws of Limits

$$\lim(f+g) = \lim f + \lim g$$

$$\lim(f-g) = \lim f - \lim g$$

$$\lim(fg) = (\lim f)(\lim g)$$

$$\lim \left(\frac{f}{g}\right) = \frac{\lim f}{\lim g}$$

provided:

$$\lim g \neq 0$$

Limit of Constant

$$\boxed{\lim_{x\to a} c = c}$$

Limit of Polynomial

For polynomial $P(x)$:

$$\boxed{\lim_{x\to a} P(x) = P(a)}$$

Limit of Rational Function

$$\lim_{x\to a} \frac{P(x)}{Q(x)} = \frac{P(a)}{Q(a)}$$

provided:

$$Q(a) \neq 0$$

Important Standard Limits

$$\boxed{\lim_{x\to 0} \frac{\sin x}{x} = 1}$$

$$\boxed{\lim_{x\to 0} \frac{1-\cos x}{x} = 0}$$

$$\boxed{\lim_{x\to 0} \frac{e^x - 1}{x} = 1}$$

$$\boxed{\lim_{x\to 0} \frac{\ln(1+x)}{x} = 1}$$

Indeterminate Forms

$$\frac{0}{0}$$

$$\frac{\infty}{\infty}$$

$$0 \cdot \infty$$

$$\infty - \infty$$

$$0^0, \quad 1^\infty, \quad \infty^0$$

Continuity

Function $f(x)$ is continuous at $x = a$ if:

$$\boxed{\lim_{x\to a} f(x) = f(a)}$$

Conditions:

$$\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = f(a)$$

Continuous Functions

Examples:

- Polynomial functions
- Rational functions
- Trigonometric functions
- Exponential functions
- Logarithmic functions

Continuity of Polynomial

All polynomial functions are continuous for all real x.

Example:

$$x^2 + 3x + 1$$

Continuity of Rational Function

Continuous wherever denominator is non-zero.
Example:

$$\frac{1}{x-2}$$

Continuous for:

$$x \neq 2$$

Continuity of Trigonometric Functions

$$\sin x, \quad \cos x$$

continuous for all real x.

$$\tan x$$

continuous where:

$$\cos x \neq 0$$

Continuity of Exponential Function

$$e^x$$

continuous for all real x.

Continuity of Logarithmic Function

$$\ln x$$

continuous for:

$$x > 0$$

Types of Discontinuity

Main types:

1. Removable discontinuity
2. Jump discontinuity
3. Infinite discontinuity
4. Oscillatory discontinuity

Removable Discontinuity

Occurs when:

$$\lim_{x \to a} f(x)$$

exists but:

$$f(a)$$

is undefined or different.
Example:

$$f(x) = \frac{x^2 - 1}{x - 1}$$

at:

$$x = 1$$

Jump Discontinuity

Occurs when:

$$\lim_{x \to a^-} f(x) \neq \lim_{x \to a^+} f(x)$$

Graph jumps suddenly.

Infinite Discontinuity

Occurs when function approaches:

$$\infty \quad \text{or} \quad -\infty$$

Example:

$$f(x) = \frac{1}{x}$$

at:

$$x = 0$$

Oscillatory Discontinuity

Occurs when function oscillates infinitely.
Example:

$$f(x) = \sin\left(\frac{1}{x}\right)$$

at:

$$x = 0$$

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Geometrical Interpretation

Continuous function can be drawn without lifting pen.

Quick Conditions

$$\text{LHL} = \text{RHL} \Rightarrow \text{Limit Exists}$$

$$\text{Limit} = f(a) \Rightarrow \text{Continuous}$$

$$\text{LHL} \neq \text{RHL} \Rightarrow \text{Discontinuous}$$

Useful Identities

$$1 - \cos x = 2\sin^2\frac{x}{2}$$

$$\sin 2x = 2\sin x\cos x$$

$$a^2 - b^2 = (a-b)(a+b)$$

Common Techniques

- Factorization
- Rationalization
- Using identities
- Substitution
- Standard limits

Matrix

Rectangular arrangement of numbers:

$$A = [a_{ij}]$$

Order:

$$m \times n$$

Types of Matrices

Row Matrix:

$$1 \times n$$

Column Matrix:

$$m \times 1$$

Square Matrix:

$$n \times n$$

Zero Matrix:
All entries zero
Identity Matrix:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Diagonal Matrix:
Non-diagonal entries zero

Equality of Matrices

Matrices are equal if:

$$a_{ij} = b_{ij}$$

for all corresponding entries.

Matrix Addition

$$A + B = [a_{ij} + b_{ij}]$$

Same order required.

Matrix Subtraction

$$A - B = [a_{ij} - b_{ij}]$$

Scalar Multiplication

$$kA = [ka_{ij}]$$

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Matrix Multiplication

If:

$$A_{m \times n}, \quad B_{n \times p}$$

then:

$$AB_{m \times p}$$

Element formula:

$$(AB)_{ij} = \sum a_{ik} b_{kj}$$

Properties of Matrix Multiplication

$$AB \neq BA$$

$$A(BC) = (AB)C$$

$$A(B + C) = AB + AC$$

Transpose of Matrix

$$A^T = [a_{ji}]$$

Interchange rows and columns.

Symmetric Matrix

$$A^T = A$$

Skew Symmetric Matrix

$$A^T = -A$$

Determinant of 2×2

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Determinant of 3×3

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

$$= a(ei - fh) - b(di - fg) + c(dh - eg)$$

Properties of Determinants

Interchanging rows changes sign.

Equal rows:

$$|A| = 0$$

Zero row:

$$|A| = 0$$

$$|A^T| = |A|$$

Minor

Minor of element a_{ij}:

$$M_{ij}$$

obtained after deleting row and column.

Cofactor

$$A_{ij} = (-1)^{i+j} M_{ij}$$

Adjoint Matrix

$$\text{Adj}(A) = [\text{Cofactors}]^T$$

Inverse of Matrix

$$A^{-1} = \frac{1}{|A|} \text{Adj}(A)$$

Condition:

$$|A| \neq 0$$

FORMULA SHEET

Inverse of 2×2

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Singular Matrix

$$|A| = 0$$

Inverse does not exist.

Non-Singular Matrix

$$|A| \neq 0$$

Inverse exists.

Cramer's Rule

For equations:

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

Main determinant:

$$\Delta = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

Formula for x

$$\Delta_x = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}$$

$$\boxed{x = \frac{\Delta_x}{\Delta}}$$

Formula for y

$$\Delta_y = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$$

$$\boxed{y = \frac{\Delta_y}{\Delta}}$$

Consistency of Equations

Unique Solution:

$$\Delta \neq 0$$

No Solution or Infinite Solutions:

$$\Delta = 0$$

Orthogonal Matrix

$$AA^T = I$$

$$A^{-1} = A^T$$

Trace of Matrix

Sum of principal diagonal elements:

$$\text{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}$$

Identity Properties

$$AI = IA = A$$

$$AA^{-1} = I$$

Important Matrix Identities

$$(A + B)^T = A^T + B^T$$

$$(AB)^T = B^T A^T$$

$$(A^{-1})^{-1} = A$$

$$(AB)^{-1} = B^{-1}A^{-1}$$

Quick Notes

$$|AB| = |A||B|$$

$$|I| = 1$$

$$|A^{-1}| = \frac{1}{|A|}$$

Applications

- Solving linear equations
- Computer graphics
- Cryptography
- Engineering problems
- Network analysis

D I F F E R E N T I A T I O N

Quick Revision Formula Sheet

Definition of Derivative

$$\boxed{f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}}$$

Basic Derivatives

$$\frac{d}{dx}(c) = 0$$

$$\frac{d}{dx}(x) = 1$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}$$

$$\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$$

Exponential and Logarithmic Functions

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(a^x) = a^x \ln a$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

$$\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$$

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Trigonometric Functions

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

Inverse Trigonometric Functions

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$$

Algebra of Derivatives

$$\frac{d}{dx}[u+v] = u' + v'$$

$$\frac{d}{dx}[u-v] = u' - v'$$

$$\frac{d}{dx}[ku] = ku'$$

Product Rule

$$\boxed{\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}}$$

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Quotient Rule

$$\boxed{\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}}$$

Chain Rule

$$\boxed{\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}}$$

Implicit Differentiation

$$\frac{d}{dx}(y) = \frac{dy}{dx}$$

Example:

$$x^2 + y^2 = 25$$

$$2x + 2y\frac{dy}{dx} = 0$$

Parametric Differentiation

$$\boxed{\frac{dy}{dx} = \frac{dy/dt}{dx/dt}}$$

Logarithmic Differentiation

$$\ln(a^n) = n \ln a$$

$$\ln(ab) = \ln a + \ln b$$

$$\ln\left(\frac{a}{b}\right) = \ln a - \ln b$$

Successive Differentiation

First derivative:

$$\frac{dy}{dx}$$

Second derivative:

$$\frac{d^2y}{dx^2}$$

Third derivative:

$$\frac{d^3y}{dx^3}$$

Tangent and Normal

Slope of tangent:

$$m = \frac{dy}{dx}$$

Equation of tangent:

$$y - y_1 = m(x - x_1)$$

Equation of normal:

$$y - y_1 = -\frac{1}{m}(x - x_1)$$

Maxima and Minima

Critical point:

$$\frac{dy}{dx} = 0$$

Minimum condition:

$$\frac{d^2y}{dx^2} > 0$$

Maximum condition:

$$\frac{d^2y}{dx^2} < 0$$

Useful Identities

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

Standard Results

$$\frac{d}{dx}(x^x) = x^x(1 + \ln x)$$

$$\frac{d}{dx}(u^v) = u^v \left(v' \ln u + v\frac{u'}{u} \right)$$

Geometrical Interpretation

Derivative represents:

$$\boxed{\text{Slope of tangent to the curve}}$$

Quick Notes

$$\frac{d}{dx}(uv) \neq u'v'$$

$$\frac{d}{dx}\left(\frac{u}{v}\right) \neq \frac{u'}{v'}$$

Differentiate term by term carefully.

I N T E G R A T I O N

Quick Revision Formula Sheet

Definition of Integration

$$\boxed{\int f(x)\,dx = F(x) + C}$$

where:

$$F'(x) = f(x)$$

Basic Integrals

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \qquad (n \neq -1)$$

$$\int \frac{1}{x}\,dx = \ln|x| + C$$

$$\int e^x\,dx = e^x + C$$

$$\int a^x\,dx = \frac{a^x}{\ln a} + C$$

FORMULA SHEET

Trigonometric Integrals

$$\int \sin x \, dx = -\cos x + C$$

$$\int \cos x \, dx = \sin x + C$$

$$\int \sec^2 x \, dx = \tan x + C$$

$$\int \csc^2 x \, dx = -\cot x + C$$

$$\int \sec x \tan x \, dx = \sec x + C$$

$$\int \csc x \cot x \, dx = -\csc x + C$$

Standard Integrals

$$\int \frac{1}{1+x^2} \, dx = \tan^{-1} x + C$$

$$\int \frac{1}{\sqrt{1-x^2}} \, dx = \sin^{-1} x + C$$

$$\int \frac{1}{x^2+a^2} \, dx = \frac{1}{a} \tan^{-1} \left(\frac{x}{a}\right) + C$$

$$\int \frac{1}{\sqrt{a^2-x^2}} \, dx = \sin^{-1} \left(\frac{x}{a}\right) + C$$

Properties of Integration

$$\int [f(x)+g(x)]dx = \int f(x)dx + \int g(x)dx$$

$$\int kf(x) \, dx = k \int f(x) \, dx$$

Substitution Method

If:

$$u = g(x)$$

then:

$$\boxed{\int f(g(x))g'(x) \, dx = \int f(u) \, du}$$

FORMULA SHEET

Integration by Parts

$$\boxed{\int uv\,dx = u\left(\int v dx\right) - \int \left[\frac{du}{dx}\left(\int v dx\right)\right] dx}$$

ILATE Rule

Priority for choosing first function:

$$I > L > A > T > E$$

- I = Inverse Trigonometric
- L = Logarithmic
- A = Algebraic
- T = Trigonometric
- E = Exponential

Partial Fractions

For rational functions:

$$\frac{P(x)}{Q(x)}$$

split into simpler fractions.
Example:

$$\frac{1}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$$

Definite Integration

$$\boxed{\int_a^b f(x)\,dx = F(b) - F(a)}$$

where:

$$F'(x) = f(x)$$

FORMULA SHEET

Properties of Definite Integrals

$$\int_a^a f(x)\,dx = 0$$

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$$

$$\int_a^b [f(x)+g(x)]dx = \int_a^b f(x)dx + \int_a^b g(x)dx$$

Important Definite Integral Results

$$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$$

If $f(x)$ is even:

$$\int_{-a}^a f(x)\,dx = 2\int_0^a f(x)\,dx$$

If $f(x)$ is odd:

$$\int_{-a}^a f(x)\,dx = 0$$

Reduction Formula

$$I_n = \int \sin^n x\,dx$$

$$I_n = -\frac{\sin^{n-1} x\cos x}{n} + \frac{n-1}{n} I_{n-2}$$

Area Under Curve

$$\boxed{\text{Area} = \int_a^b y\,dx}$$

Average Value of Function

$$\boxed{f_{\text{avg}} = \frac{1}{b-a}\int_a^b f(x)\,dx}$$

Improper Integrals

Infinite limits:

$$\int_a^\infty f(x)\,dx$$

Discontinuous integrals:

$$\int_0^1 \frac{1}{\sqrt{x}}dx$$

Useful Identities

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

Common Integrals

$$\int \tan x\,dx = \ln|\sec x| + C$$

$$\int \cot x\,dx = \ln|\sin x| + C$$

$$\int \sec x\,dx = \ln|\sec x + \tan x| + C$$

$$\int \csc x\,dx = \ln|\csc x - \cot x| + C$$

Quick Notes

$$\int f'(x)\,dx = f(x) + C$$

Integration is reverse process of differentiation.

Always add constant of integration:

$$+C$$

APPLICATIONS OF DERIVATIVES

Quick Revision Formula Sheet

Slope of Tangent

Derivative gives slope of tangent.

$$\boxed{m = \frac{dy}{dx}}$$

Equation of tangent:

$$y - y_1 = m(x - x_1)$$

Slope of Normal

Slope of normal:

$$\boxed{m_n = -\frac{1}{m}}$$

Equation of normal:

$$y - y_1 = -\frac{1}{m}(x - x_1)$$

Increasing Function

Function is increasing if:

$$\boxed{\frac{dy}{dx} > 0}$$

Decreasing Function

Function is decreasing if:

$$\boxed{\frac{dy}{dx} < 0}$$

Critical Points

Critical points occur when:

$$\boxed{\frac{dy}{dx} = 0}$$

or derivative is undefined.

Maximum Point

For maxima:

$$\frac{dy}{dx} = 0$$

and

$$\boxed{\frac{d^2y}{dx^2} < 0}$$

Minimum Point

For minima:

$$\frac{dy}{dx} = 0$$

and

$$\boxed{\frac{d^2y}{dx^2} > 0}$$

Point of Inflection

Occurs when:

$$\boxed{\frac{d^2y}{dx^2} = 0}$$

and concavity changes.

Concavity

Concave Up:

$$\frac{d^2y}{dx^2} > 0$$

Concave Down:

$$\frac{d^2y}{dx^2} < 0$$

FORMULA SHEET

Rate of Change

Average rate:

$$\frac{\Delta y}{\Delta x}$$

Instantaneous rate:

$$\boxed{\frac{dy}{dx}}$$

Velocity

If:

$$s = f(t)$$

then velocity:

$$\boxed{v = \frac{ds}{dt}}$$

Acceleration

Acceleration:

$$\boxed{a = \frac{dv}{dt} = \frac{d^2 s}{dt^2}}$$

Related Rates

If variables depend on time:

$$\boxed{\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}}$$

Approximation

Using linear approximation:

$$\boxed{f(x + \Delta x) \approx f(x) + f'(x)\Delta x}$$

Differentials

$$dy = f'(x)dx$$

Approximate change:

$$\Delta y \approx dy$$

FORMULA SHEET

Elasticity

$$\boxed{E = \frac{x}{y}\frac{dy}{dx}}$$

Marginal Cost

If cost function is $C(x)$:

$$\boxed{MC = \frac{dC}{dx}}$$

Marginal Revenue

If revenue function is $R(x)$:

$$\boxed{MR = \frac{dR}{dx}}$$

Marginal Profit

If profit function is $P(x)$:

$$\boxed{MP = \frac{dP}{dx}}$$

Optimization Problems

Steps:

1. Form function
2. Differentiate
3. Set derivative zero
4. Use second derivative test

Curve Sketching

Use derivatives to determine:

- Increasing intervals
- Decreasing intervals
- Maxima
- Minima

- Concavity
- Inflection points

Newton-Raphson Formula

$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}}$$

Geometrical Interpretation

Derivative represents:

$$\boxed{\text{Slope of tangent to curve}}$$

Second derivative represents:

$$\boxed{\text{Concavity of curve}}$$

Useful Conditions

Horizontal tangent:

$$\frac{dy}{dx} = 0$$

Vertical tangent:

$$\frac{dx}{dy} = 0$$

Motion Formulas

$$v = \frac{ds}{dt}$$

$$a = \frac{dv}{dt}$$

$$a = \frac{d^2s}{dt^2}$$

Quick Notes

$$f'(x) > 0 \Rightarrow \text{Increasing}$$

$$f'(x) < 0 \Rightarrow \text{Decreasing}$$

$$f''(x) > 0 \Rightarrow \text{Minimum}$$

$$f''(x) < 0 \Rightarrow \text{Maximum}$$

A P P L I C A T I O N S O F I N T E G R A T I O N

Quick Revision Formula Sheet

Area Under Curve

Area bounded by curve:

$$\boxed{A = \int_a^b y\,dx}$$

where:

$$y = f(x)$$

Area Between Two Curves

If upper curve is $f(x)$ and lower curve is $g(x)$:

$$\boxed{A = \int_a^b [f(x) - g(x)]dx}$$

Area with Respect to y

If:

$$x = f(y)$$

then:

$$\boxed{A = \int_c^d x\,dy}$$

FORMULA SHEET

Area in Polar Coordinates

For polar curve:

$$r = f(\theta)$$

Area:

$$\boxed{A = \frac{1}{2}\int_{\alpha}^{\beta} r^2\, d\theta}$$

Volume of Solid of Revolution

About x-axis:

$$\boxed{V = \pi \int_a^b y^2\, dx}$$

Volume About y-axis

$$\boxed{V = \pi \int_c^d x^2\, dy}$$

Volume by Cylindrical Shells

$$\boxed{V = 2\pi \int_a^b xy\, dx}$$

Surface Area of Revolution

About x-axis:

$$\boxed{S = 2\pi \int_a^b y\sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx}$$

Average Value of Function

$$\boxed{f_{\text{avg}} = \frac{1}{b-a}\int_a^b f(x)\, dx}$$

FORMULA SHEET

Displacement

If velocity is $v(t)$:

$$\boxed{s = \int v(t)\,dt}$$

Distance Travelled

$$\boxed{\text{Distance} = \int |v(t)|\,dt}$$

Velocity from Acceleration

$$\boxed{v = \int a(t)\,dt}$$

Position from Velocity

$$\boxed{s = \int v(t)\,dt}$$

Work Done

If force is variable:

$$\boxed{W = \int_a^b F(x)\,dx}$$

Pressure Force

$$\boxed{F = \int p(y)\,dA}$$

Center of Mass

$$\boxed{\bar{x} = \frac{\int_a^b x f(x)\,dx}{\int_a^b f(x)\,dx}}$$

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Arc Length

$$\boxed{L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx}$$

Probability Density Function

If $f(x)$ is PDF:

$$\boxed{P(a \le x \le b) = \int_a^b f(x)\, dx}$$

Consumer Surplus

$$\boxed{CS = \int_0^q D(x)\, dx - pq}$$

Producer Surplus

$$\boxed{PS = pq - \int_0^q S(x)\, dx}$$

Improper Integrals

Infinite interval:

$$\boxed{\int_a^\infty f(x)\, dx}$$

Discontinuous integrand:

$$\boxed{\int_0^1 \frac{1}{\sqrt{x}} dx}$$

Definite Integral Formula

$$\boxed{\int_a^b f(x)\, dx = F(b) - F(a)}$$

where:

$$F'(x) = f(x)$$

FORMULA SHEET

Useful Properties

$$\int_a^a f(x)\,dx = 0$$

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$$

$$\int_a^b [f(x) + g(x)]dx = \int_a^b f(x)dx + \int_a^b g(x)dx$$

Even and Odd Functions

If $f(x)$ is even:

$$\boxed{\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx}$$

If $f(x)$ is odd:

$$\boxed{\int_{-a}^{a} f(x)\,dx = 0}$$

Important Results

$$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$$

$$\frac{d}{dx}\left(\int_a^x f(t)\,dt\right) = f(x)$$

Quick Notes

Integration is used for:

- Area
- Volume
- Distance
- Work
- Probability
- Average value
- Arc length
- Surface area